

Task-1. Pick a use case from an app you use daily (for example: Instagram photo sharing, Zomato food ordering, or Flipkart online shopping) and briefly describe the main technical requirements for hosting this app online.

Use Case: Zomato Food Ordering

- **Compute:** Servers to run the Zomato website, mobile app backend, and APIs.
- **Database:** To store user accounts, restaurant details, menus, orders, and delivery information.
- **Storage:** To store restaurant and food images.
- **Networking:** Load balancers and a reliable network to handle millions of users.
- **Security:** Authentication, encryption, firewalls, and access control to protect user and payment information.
- **Scalability:** Auto-scaling to handle sudden increases in orders during peak hours.
- **Monitoring and backup:** Monitoring services to detect failures and regular backups to prevent data loss.

Task-2. Create a table comparing public, private, and hybrid cloud deployment models for your chosen use case. Include at least 4 factors: cost, security, scalability, and ease of maintenance. Think about how each model would handle user data, sudden traffic spikes, and updates for your app scenario.

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Cost	Lower initial cost because resources are rented and paid for based on usage.	Higher cost because the company must maintain its own infrastructure.	Medium to high because both public and private infrastructure are required.
Security	Provides strong security tools, but Zomato must configure them correctly.	Provides greater control over infrastructure and sensitive information.	Sensitive data can be kept in the private environment while other services use the public cloud.

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Scalability	Excellent; resources can quickly increase during sudden order spikes.	Scaling may require purchasing and installing additional hardware.	Very good; public-cloud resources can handle extra traffic when private resources are full.
Ease of maintenance	Easier because the cloud provider manages much of the infrastructure.	More difficult because Zomato has to manage its own infrastructure.	More complex because both environments must be managed and connected.
Traffic spikes	Handles sudden increases well through auto-scaling.	May struggle if enough capacity is not available.	Public cloud can absorb additional traffic during peak periods.
Data management	Data can be stored securely using encryption and access controls.	Provides maximum control over where sensitive data is stored.	Allows sensitive data to remain private while less-sensitive workloads use public cloud.

Task-3. Write a short paragraph explaining which cloud deployment model you would recommend for your chosen use case and why. Support your answer with at least two points from your comparison table.

I would recommend the **Hybrid Cloud** model for Zomato. It provides good **security and control** because sensitive customer and business information can be kept in a private environment. At the same time, Zomato can use the **scalability of the public cloud** to handle sudden increases in orders during busy periods such as weekends, festivals, or special offers. Although hybrid cloud is more complex to maintain, its combination of security and scalability makes it suitable for a large food-delivery application.

Task-4. Imagine your chosen app becomes extremely popular overnight (like during IPL finals or a big sale). Describe one advantage and one disadvantage of using a hybrid cloud model in this situation.

Zomato Becomes Extremely Popular Overnight

Advantage

The biggest advantage of a **hybrid cloud** is **scalability**. During a sudden rush, such as a major festival or promotional sale, Zomato can use public-cloud resources to handle the extra users and orders without immediately purchasing additional physical servers.

Disadvantage

The main disadvantage is **complexity**. Managing and connecting private and public cloud environments can be difficult and expensive. A networking or configuration problem between the two environments could affect the application's performance.

Task-5. Use ChatGPT or any AI tool to generate a summary of when to use public, private, and hybrid clouds for popular Indian apps (such as Paytm, IRCTC, BookMyShow). Paste the summary and highlight one insight you found surprising

App	Recommended Model	When it is useful
Paytm	Hybrid Cloud	Suitable when the application needs strong protection and control over sensitive financial/customer information while also requiring scalability for large numbers of transactions.
IRCTC	Hybrid Cloud	Useful because railway booking systems can contain critical information that needs strong control, while public-cloud resources can help handle massive booking traffic during peak periods.
BookMyShow	Public Cloud	Suitable for handling unpredictable traffic spikes when popular movies, concerts, or IPL tickets go on sale. Public-cloud resources can scale according to demand.